

From Black-Box to Glass-Box: A User-Centric Framework for Explainable Visualization with LLMs

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Abstract—Large Language Models (LLMs) have revolutionized the fields of Artificial Intelligence and Data Science, holding great potential in a variety of domains such as E-Learning, Business, and Data Analytics. However, the challenge of relying on the output generated by LLMs still persists, specifically in sensitive domains such as healthcare and finance. This paper presents an enhanced framework for generating explainable visualizations using LLMs as well as introducing an approach to make the LLM outputs more interpretable to users, enhancing their trust in the validity of the output. The work presented in this paper builds upon the capabilities of the existing tool: LIDA [7]. The paper proposes modifications to its core components as well as the addition of new components. The goal is to enhance the explainability and reliability of the produced outputs. The proposed system refines how goals are identified and explored as well as how visualizations are assessed. The aim is to address limitations in explainability. Key contributions of this work include the integration of a new scoring mechanism that ensures transparency for the LLM-generated evaluations across multiple dimensions, enforcing user trust in the outputs generated. This study contributes by merging the growing field of explainable Artificial Intelligence with the domain of data exploration and analytics by offering a modular and extensible approach to visualization generation, emphasizing user-centric design and interpretability. The findings have implications for researchers, data analysts, and developers looking to bridge the gap between complex data and actionable insights.

I. INTRODUCTION

Visual representations of data is powerful impact if used correctly and efficiently. They can communicate insights in raw data that can meet a business need or even improve a business model or the understanding of a program [23], [24], [26]. Visualizations build a relationship between the viewer and the raw data and thus generating quality visuals tailored to a dataset is an area of large interest specifically in the domain of business intelligence where data visualization is a crucial cornerstone. Data visualization also serves an impactful role in understanding and interpreting the output of machine learning models [1] as shown in popular model-agnostic interpretability approaches such as LIME [21] and SHAP [16] which rely heavily on enhancing the explainability of black-box models.

Previous work towards visualization recommendation systems were rule-based systems with heuristics crafted by domain experts, [18], [25], this approach worked but was difficult in terms of maintainability and the task of crafting

a general set of rules was difficult as each dataset is unique. [20]

Other approaches were Machine learning based towards visualization recommendation [8], [13], [17], [20] which was a better approach in contrast to rule-based systems. The main drawback was the intensive training time for the model, bias and the output visualizations were the product of a "black-box" model therefore the results were not interpretable by nature.

Large Language Models have demonstrated clear advances in many fields, including data visualization. Where one of the most notable advancements is granting users the ability to generate charts using natural language commands. This makes the domain of data visualization more accessible to users with low technical abilities.

The rapid growth in LLM usage by users with diverse needs and goals presents a significant challenge regarding reliability. Before acting on or utilizing LLM outputs in sensitive domains such as healthcare, finance, and data visualization, users need to question the reasoning behind the output.

This concern is particularly prominent in the area of visualization recommendations, where the charts produced by LLMs must not only align with user intent but also convey information that is meaningful and suitable for presentation to stakeholders. Moreover, current LLMs function as black-box models, providing no clear evidence or insight into the underlying mechanisms that lead to a specific output.

In this work, we build upon LIDA [7], a tool designed for the automatic generation of grammar-agnostic visualizations and infographics using LLMs. Our contributions focus on enhancing the interpretability of the recommended charts and improving the evaluation mechanisms to ensure their reliability and alignment with user needs.

II. LITERATURE REVIEW

The black-box approach of different machine learning models are always a challenge for AI adoption. Even though the models are becoming more accurate, the internal decision-making processes are becoming more opaque. This reduces the confidence of users in predictions [12]. Explainable AI (XAI) generates justifications for model decisions. These explanations may be intrinsic—produced by inherently interpretable models—or post-hoc, added after training to make outputs

more intelligible [2], [3], [6]. A widely accepted framework categorizes XAI approaches by four key dimensions which are local versus global explanation scope, model-specific versus model-agnostic methods, intrinsic versus post-hoc generation, and the format of explanation (e.g., textual summaries or visual feature importance maps) [4], [5], [29].

Another issue is usually the trade-off between interpretability and predictive performance [10], [22]. Highly interpretable models are often constrained in their complexity and accuracy. High-performing models such as neural networks are typically less transparent. Tools like LIME (Local Interpretable Model-Agnostic Explanations) [21] address this by perturbing input data around an instance of interest as well as training a simple surrogate model to approximate the behavior of the complex model locally. LIME is especially effective at explaining single predictions but does not provide insight into global model behavior.

SHAP (SHapley Additive exPlanations) [16] offers a more theoretically grounded solution based on cooperative game theory. This assigns each feature an importance score by averaging over all possible feature combinations. While SHAP is model-agnostic and satisfies several desirable mathematical properties, its computation cost increases significantly with model complexity and input dimensionality.

Prospector introduced in [15] presents an interactive visual analytics tool that allows users to explore model predictions using partial dependence plots. The plots show the marginal effect of a single feature while averaging over others. This gives users a high-level sense of feature influence. However, Prospector focuses mainly on binary classification. However, it struggles with scalability and multi-class prediction tasks.

There are also some issues with forward-model XAI methods like LIME and SHAP for producing unintuitive explanations. AIME (Approximate Inverse Model Explanations) [19] was proposed as a more intuitive alternative. It constructs approximate inverse mappings. The aim is to estimate what input features would lead to a given output. The framework enables both local and global interpretability along with visualizations. Such visualizations represent similarity distributions across data instances. AIME is promising. However, its success depends on mathematical conditions such as matrix non-singularity. This limits its generalizability in highly complex models.

The domain of visualization recommendation systems aims at simplifying data exploration by automating chart selection and generation [1]. Traditional systems such as Voyager and Voyager 2 [27], [28] are rule-based. They use handcrafted heuristics to guide chart selection. Rule-based systems lack adaptability and struggle with novel datasets. But they are effective in constrained environments.

Machine learning-based systems have emerged as more scalable alternatives. VizML [13] was one of the earliest models to frame visualization recommendation as a prediction task, using a large dataset of visualization design choices to learn patterns. However, VizML recommends design choices

rather than full visualizations, limiting its usefulness for non-experts.

To address this, Qian et al. [20] proposed a wide and deep learning model capable of recommending complete visualizations. The system ranks candidate visualizations using an enhanced nDCG metric that considers both relevance and presentation quality.

III. LIDA++

LIDA++ is a user-centric platform. It improves explainability as well as usability in AI-driven visualization systems. It extends the original LIDA framework [7]. The updated modules are designed for addressing intrinsic deficits within explainability and evaluation. The platform is designed for seamless interaction, guiding users through a structured tour on first access. The backend, developed in Python using FastAPI, communicates efficiently with a responsive Next.js frontend.

The system architecture, shown in Figure 1, highlights several key modifications. Goal Explorer and the Goal Analyzer constitute a duo of separate modules into which LIDA++ divides the initial goal exploration segment. This separation enables more transparent and modular interactions. Additionally a novel visualization evaluator module was introduced, since it rectifies LIDA's prior incapability regarding appraisal of the chart output. Furthermore, the summarization component has been optimized to reduce token usage. This significantly improves performance and stability when working with high-dimensional datasets.

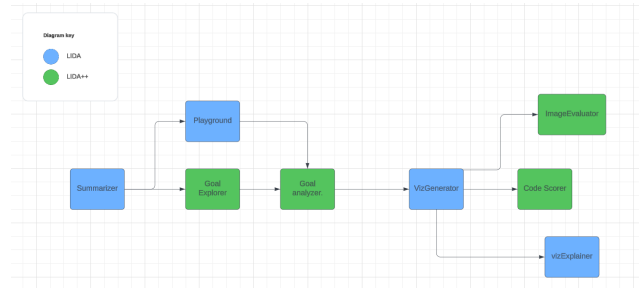


Fig. 1: LIDA++ system architecture and key modules.

The landing module, introduces the components and navigation paths of the system. This component emphasizes accessibility, ensuring that novice and expert users alike can understand and navigate the platform.

Users can upload datasets using the data summary module. LIDA++ then generates enriched summaries using LLMs, incorporating both general descriptors and selected statistical properties. LIDA++ divides summaries into two parts unlike LIDA. Thus, it send the general properties for enrichment. This optimization reduces token usage. It also improves response times, and minimizes hallucinations. For instance, while the standard LIDA system failed to process a dataset with 130 features, LIDA++ handled it successfully, demonstrating improved scalability.

Goal generation in LIDA++ begins with the Goals module, where users can choose from a curated list of data or enter free-text queries. Figure 2 displays this interface. Goals are generated using the Goal Explorer and further analyzed through the Goal Analyzer. This modular breakdown enables more targeted and context-aware recommendations. LIDA returns a bundled set of goals, visualizations, and rationales. However, LIDA++ allows users to choose a goal. The analyzer then contextualizes by detecting involved features, data types, and visualization intent. This results in an analysis document explaining the rationale behind the recommendation to enhance interpretability.

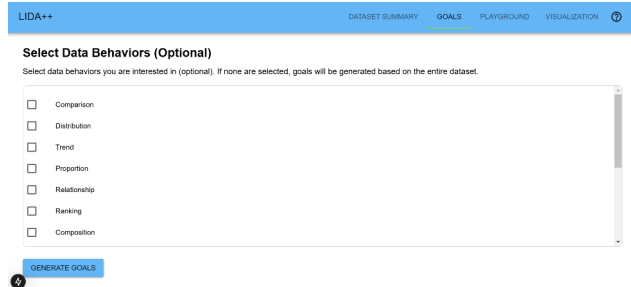


Fig. 2: Goal selection and exploration interface.

The Playground module (Figure 3) provides a space for querying datasets with customized goals. Users are guided through best practices such as including dataset feature names or specifying the desired chart type. This helps the system generate better recommendations. If a user specifies a goal like “What is the relationship between X and Y using a scatter plot?”, LIDA++ uses the Goal Analyzer to return a recommended chart and detailed rationale. This is presented prior to execution. This interactive and explainable recommendation process improves transparency compared to the original LIDA, where recommendations were executed without elaboration.

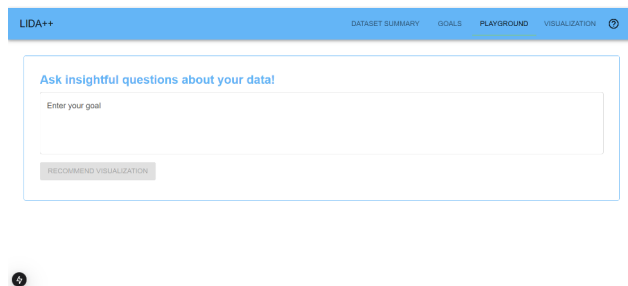


Fig. 3: Playground interface for custom goal submission.

Once a goal is selected, users are redirected to the Visualization module (Figure 4). This shows the generated chart alongside the Python code used to create it. It includes evaluation buttons that allow users to assess both code and chart quality. The original LIDA evaluation used the Self-Evaluated Visualization Quality (SEVQ) metric [7]. This relied on LLMs to score outputs based on six dimensions. However, inconsistencies were observed: identical rationales sometimes

produced different scores. LIDA++ addresses this by modifying SEVQ to use a structured logic-based approach.

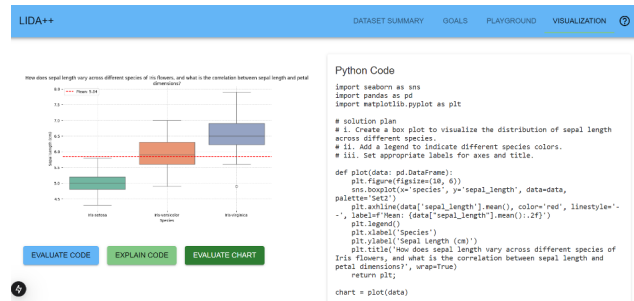


Fig. 4: Visualization module for rendering and interacting with charts.

LIDA++ introduces the ImageEvaluator module to address the limitation in evaluating the rendered chart. This component uses multi-modal LLMs capable of analyzing both text and image inputs. It evaluates charts across five dimensions: clarity, accuracy, insightfulness, design, and compliance. This dual-layer evaluation—code and visual output—offers a more reliable and comprehensive quality assessment.

The modified SEVQ process, shown in Figure 5, asks the LLM to respond to a predefined set of questions using a structured yes/no format, with rationale and point allocation. Each question is categorized as either default, reversed, or conditional—ensuring that scoring logic aligns with the intended meaning. For instance, a reversed question like “Are there any syntax errors?” earns points for “no” answers. But a conditional question might grant points by default if its premise does not apply. This method corrects prior inconsistencies and enhances user trust in the evaluation process.

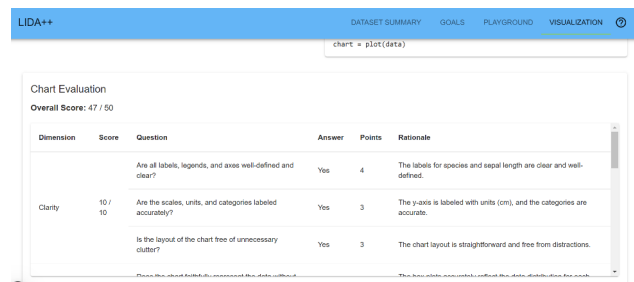


Fig. 5: Sample output from chart evaluation using the ImageEvaluator.

IV. TESTING

To evaluate the effectiveness, usability, and explainability of LIDA++, a multi-phase testing process was conducted. It incorporated both qualitative and quantitative methods. The evaluation began with a focus group involving six computer science professionals specializing in software engineering, data science, and IT security. Their feedback emphasized the need for enhanced navigation features. This includes

The screenshot shows the 'Code Evaluation' section of the LIDA++ interface. It displays a table with columns: Dimension, Score, Question, Answer, Points, and Rationale. The overall score is 60 / 60. The table lists several evaluation questions related to syntax, logic, variable names, and imports/variables, all of which were answered correctly (Yes/No) and awarded points.

Dimension	Score	Question	Answer	Points	Rationale
		Are there syntax errors that prevent the code from running?	No	4	The code has no syntax errors and runs successfully.
		Are there logic errors that lead to incorrect outputs?	No	3	The logic appears sound for the intended visualization.
Bugs	10 / 10	Are variable names used consistently?	Yes	1	Variable names such as 'data' and 'chart' are used consistently.
		Are there unused imports or variables?	No	1	All imports and variables are utilized in the code.

Fig. 6: Code evaluation output using the modified SEVQ scoring logic.

embedded website tours and contextual tooltips. This aims to help users understand the functionality of each module and maintain orientation within the system. Additionally, participants recommended improvements in frontend error handling, particularly in the form of clear and actionable error messages to guide users during unexpected events or invalid inputs.

A structured questionnaire was developed to assess the core components of LIDA++' specifically the data summary, goal generation, evaluation mechanisms, and the overall explainability of the system. The questionnaire was designed to capture user perspectives on satisfaction, trust, and clarity across all interactive modules. It included sections for demographic information (age, expertise, professional background), followed by targeted questions addressing the clarity and relevance of dataset summaries, the quality and complexity of the generated goals, the transparency of evaluation scores, and the interpretability of code and chart explanations. To ensure methodological rigor, the questionnaire incorporated standardized evaluation tools such as the System Usability Scale (SUS) [11] and NASA Task Load Index (TLX) [14], enriching the insights obtained and supporting cross-study comparability.

To systematically compare LIDA++ with the original LIDA, the same evaluation instrument was used for both systems. After participants had completed separate sessions with each system, a comparative questionnaire was administered to elicit user preferences, perceived differences in usability and clarity, and overall system satisfaction. This final step enabled direct comparison between the two platforms and identified areas of relative strength and weakness. The original LIDA interface used during testing was the publicly available implementation provided by Microsoft [7], ensuring consistency and reproducibility of results across participants.

The experimental design focused on fairness, engagement, and consistency. The LLM model used for both systems was OpenAI's GPT-4, selected for its superior performance in generating coherent code and natural language explanations. Each participant was presented with a set of five curated datasets and asked to choose one based on personal interest. This design ensured user engagement while maintaining consistency; once a dataset was selected, it was used across both systems to enable fair comparison of outputs such as goals, visualizations,

and generated explanations.

Participants were instructed to maintain a critical lens throughout the evaluation, particularly with regard to clarity and completeness of the system's explanations. After uploading a dataset, users reviewed the generated summary and assessed its readability and informativeness. They then evaluated the system-generated goals and selected one of interest to explore further. After selecting a goal, participants examined the corresponding visualization and underlying code, using the "Explain Code" functionality to assess the rationale provided. Finally, users explored the system's evaluation tools, analyzing the generated scores and explanations for clarity and usefulness. To complete the evaluation, participants submitted their own custom goal query and judged whether the system provided an intelligible explanation for the recommendation.

To minimize bias from the order in which systems were used, 24 participants were randomly divided into two counterbalanced clusters, as shown in Table I. Cluster 1 began with LIDA++, followed by its survey, then used LIDA, and completed a final comparison survey. Cluster 2 followed the reverse order—starting with LIDA, then LIDA++, and concluding with a second version of the comparison survey.

This counterbalanced setup ensured an impartial comparison of both systems and reduced potential sequence bias. It also enabled additional observations about whether prior exposure to LIDA++ facilitated easier or more insightful interaction with LIDA, or vice versa. By triangulating focus group insights, quantitative survey results, and direct system comparisons, the evaluation methodology provided a comprehensive and user-centered assessment of LIDA++'s capabilities.

V. ANALYSIS OF USER RESPONSES

This section presents a comprehensive analysis of the results obtained from both participant clusters, focusing on survey feedback, task performance, and system perceptions. The analysis reveals comparative insights into LIDA++ and LIDA across usability, clarity, workload, and trust dimensions.

A. Cluster 1 Overview

Participants in Cluster 1 primarily had a strong technical background: 75% held bachelor's degrees, 50% had expertise in data science, and all (100%) had prior experience with LLM-based tools. This technically proficient group was well-equipped to critically evaluate both systems.

1) *Summary Perception*: Figures 8 to 9 illustrate that LIDA++ was consistently rated higher in summary clarity, informativeness, and the ability to detect outliers. Participants strongly preferred LIDA++ for conveying essential dataset insights.

2) *Goal Insights*: Figures 7 to 9 in the Goals section indicate that participants found LIDA++'s goals more insightful, actionable, and clear. These goals effectively guided further exploration and were frequently used as a reference in the playground module.

Cluster	Step 1	Step 2	Step 3
1	LIDA++ → Survey	LIDA → Survey	Final Comparison Survey 1
2	LIDA → Survey	LIDA++ → Survey	Final Comparison Survey 2

TABLE I: Flow of tasks and surveys for both participant clusters.

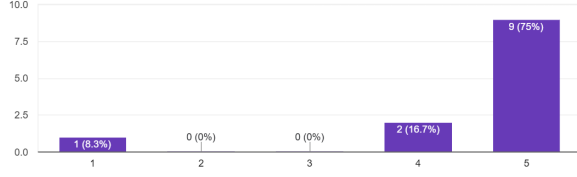


Fig. 7: LIDA++

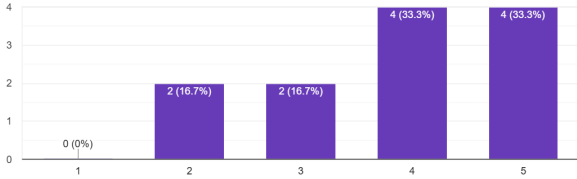


Fig. 8: LIDA

Fig. 9: Summary clarity comparison between LIDA++ and LIDA in Cluster 1

3) *Evaluation Comprehension and Trust*: Figures 7 to 9 reveal that LIDA++’s evaluation framework was better understood and more trusted. The scoring logic was perceived as transparent, and participants expressed high confidence in both code and chart evaluations.

4) *Explainability Metrics*: Figures 7 to 9 show that LIDA++ provided more consistent justifications for goals and visualization choices. Participants trusted LIDA++ outputs more and rated it higher on overall explainability.

5) *Workload (NASA-TLX)*: Across six dimensions (mental demand, effort, performance, etc.), LIDA++ required less effort and induced less frustration than LIDA. Table II and III summarize the scores, confirming a lower perceived workload with LIDA++.

Dimension	LIDA++C1	LIDAC1	LIDA++C2	LIDAC2
Mental Demand	3.17	3.83	2.92	2.92
Physical Demand	2.08	2.75	1.75	1.83
Temporal Demand	3.08	3.17	2.17	2.92
Performance	6.42	5.58	5.92	5.17
Effort	2.58	3.17	2.58	3.08
Frustration	2.08	2.50	2.17	2.75

TABLE II: Average NASA-TLX Scores Across Clusters

6) *Usability (SUS)*: Cluster 1 reported an SUS score of 81.25 for LIDA++, classified as "Excellent", while LIDA received a lower score of 62.08, categorized as "Poor". This reinforces the overall user preference and positive experience with LIDA++.

Dimension	LIDA++	LIDA
Mental Demand	3.04	3.38
Physical Demand	1.92	2.29
Temporal Demand	2.62	3.04
Performance	6.17	5.38
Effort	2.58	3.12
Frustration	2.12	2.62

TABLE III: Average NASA-TLX Scores for LIDA++ and LIDA Unified

B. Cluster 2 Overview

Cluster 2 consisted of more diverse users, including 41.7% from software engineering and 66.7% with prior LLM experience. However, 66.7% were beginners with visualization tools, representing a less technically advanced user group.

1) *Summary Perception*: As shown in Figures 7 to 9, participants appreciated LIDA++ for its clearer summaries and better contextual explanations. LIDA++ enabled users to understand dataset context with minimal additional support.

2) *Goal Insights*: LIDA++ was again favored for generating goals that were easier to understand, more insightful, and actionable (see Figures 7 to 9). This affirmed its ability to guide novice users effectively.

3) *Evaluation Trust*: Participants expressed greater trust in LIDA++’s evaluation mechanism, especially for visualizations (Figure 9). LIDA’s lack of this feature further widened the gap.

4) *Explainability Metrics*: Figures 7 to 9 show that LIDA++ provided sufficient and reliable justifications for both goal and chart recommendations, increasing user confidence across outputs.

5) *Workload (NASA-TLX)*: Cluster 2 showed reduced workload when using LIDA++, especially in mental demand, effort, and frustration. These results, found in Table II, underscore LIDA++’s effectiveness for novice users.

6) *Usability (SUS)*: LIDA++ received a SUS score of 77.71 in Cluster 2, rated "Good", compared to LIDA’s 59.17, which again fell into the "Poor" tier. Despite lower technical familiarity, users appreciated the system’s design and clarity.

C. Comparative and Statistical Findings

Both qualitative and quantitative comparisons (Figures 7 to 9) show that LIDA++ consistently outperformed LIDA across metrics like trust, clarity, goal quality, and overall satisfaction. 75% of Cluster 1 and 91.7% of Cluster 2 preferred LIDA++.

D. Inferential Statistics

Usability (SUS): A Wilcoxon signed-rank test yielded $p = 0.0039$, rejecting the null hypothesis. LIDA++ had a mean score of 79.5 vs. LIDA’s 60.6.

Goals: Wilcoxon signed-rank test, $p = 0.0139$, confirmed significantly better goal clarity and relevance for LIDA++ (mean = 12.63) over LIDA (mean = 10.96).

Evaluation: With $p = 0.0001$, evaluation scores favored LIDA++ (mean = 8.54) over LIDA (mean = 5.88), affirming the impact of the explainable scoring framework.

Explainability: A paired t-test revealed $p < 0.000001$ (mean = 17.42 for LIDA++, 11.83 for LIDA), affirming the strength of LIDA++ in producing understandable and justifiable outputs.

Order Effects: SUS scores were not significantly affected by presentation order ($p = 0.96$), validating the study's design and allowing safe aggregation of results.

VI. CONCLUSIONS AND FUTURE WORK

LIDA++ was introduced to extend explainability for LIDA which already had very strong basis. Improvements to the summarizer enhanced usability with wide datasets, and the new evaluation framework increased transparency, making LIDA++ more reliable, especially in sensitive domains. Despite these strengths, LIDA++ faced limitations, including occasional faulty code generation by the LLM and some inaccuracies in evaluation scores. Future work should explore LLM fine-tuning with documentation, user-defined scoring criteria, code tracing for line-level explainability, and enhanced visual data summaries for better initial dataset understanding. The collaboration and extended reality visualization dimension could be also explored in the future [9].

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